

Lecture 5

Membrane transport



Dr Philip Button



Transport of nutrients

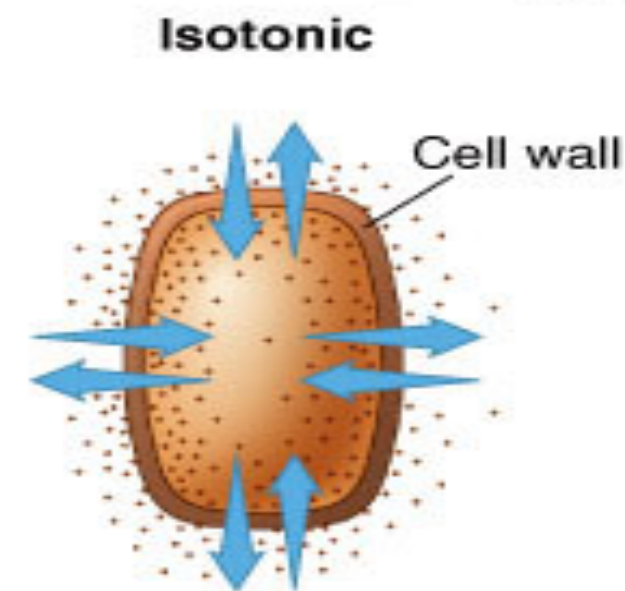
- Osmosis
- Diffusion
- Active transport
- Endocytosis
 - Pinocytosis
 - Phagocytosis



Osmosis

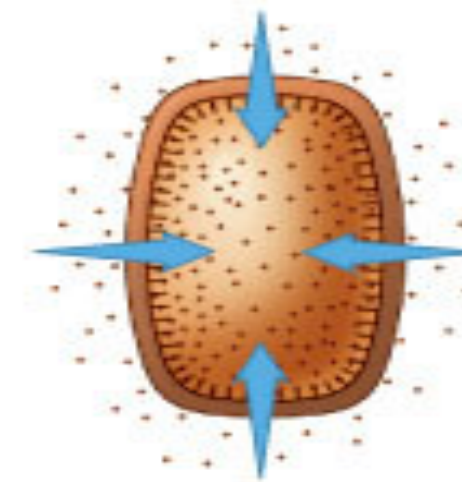
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Cells with Cell Wall



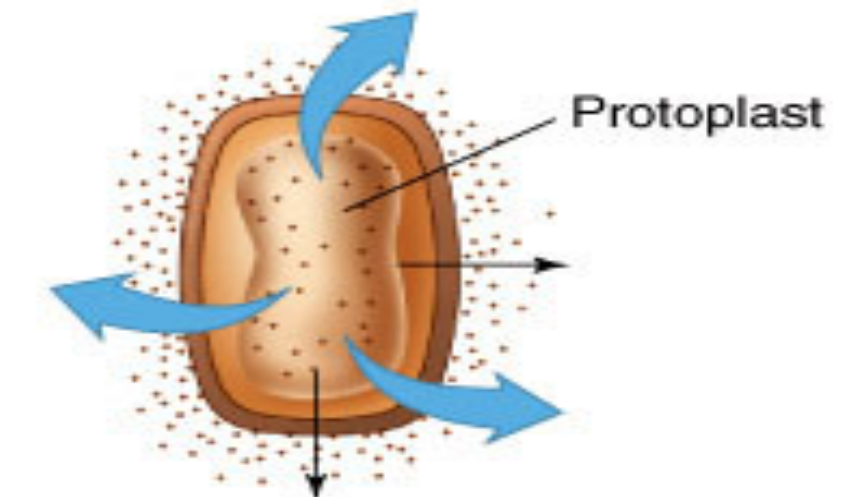
Water concentration is equal inside and outside the cell, thus rates of diffusion are equal in both directions.

Hypotonic



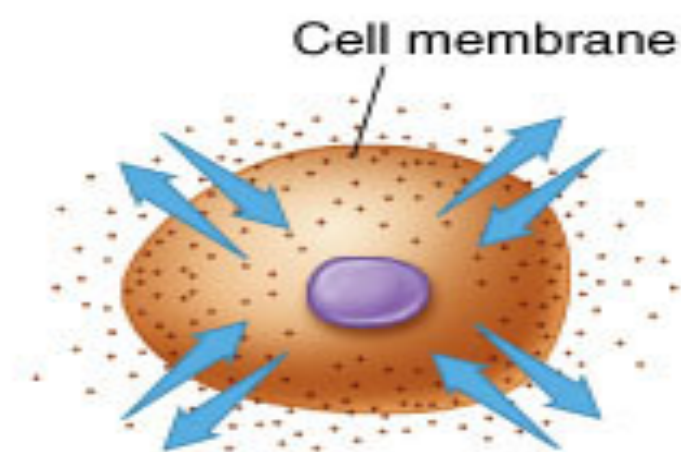
Net diffusion of water is into the cell; this swells the protoplast and pushes it tightly against the wall. Wall usually prevents cell from bursting.

Hypertonic

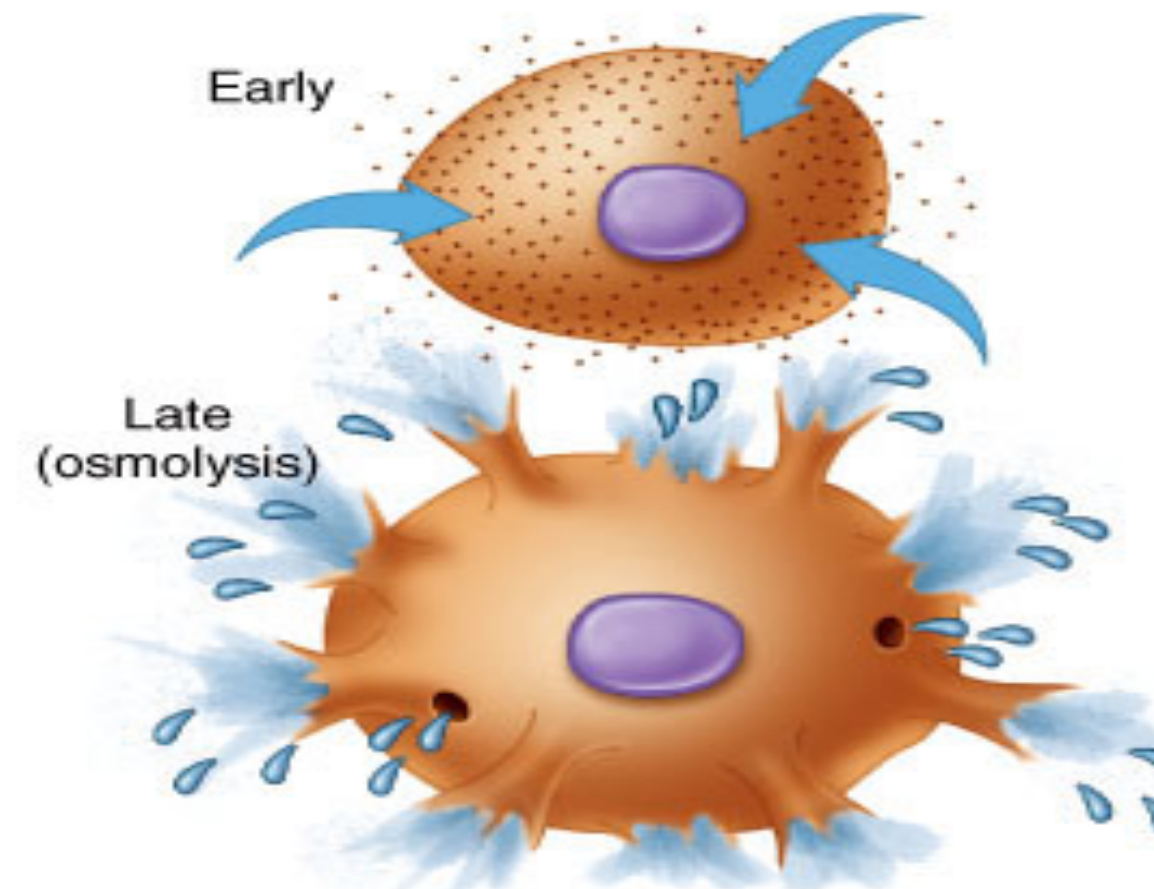


Water diffuses out of the cell and shrinks the protoplast away from the cell wall; process is known as **plasmolysis**.

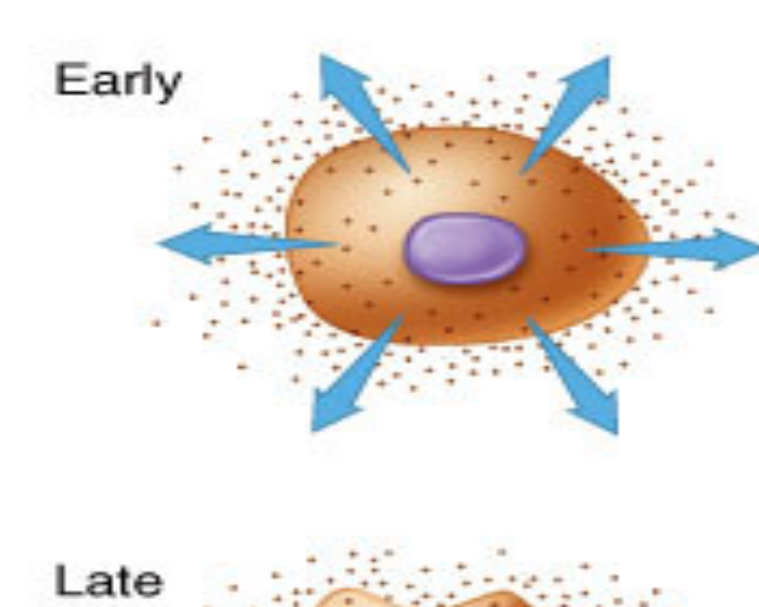
Cells Lacking Cell Wall



Rates of diffusion are equal in both directions.



Diffusion of water into the cell causes it to swell, and may burst it if no mechanism exists to remove the water.



Water diffusing out of the cell causes it to shrink and become distorted.

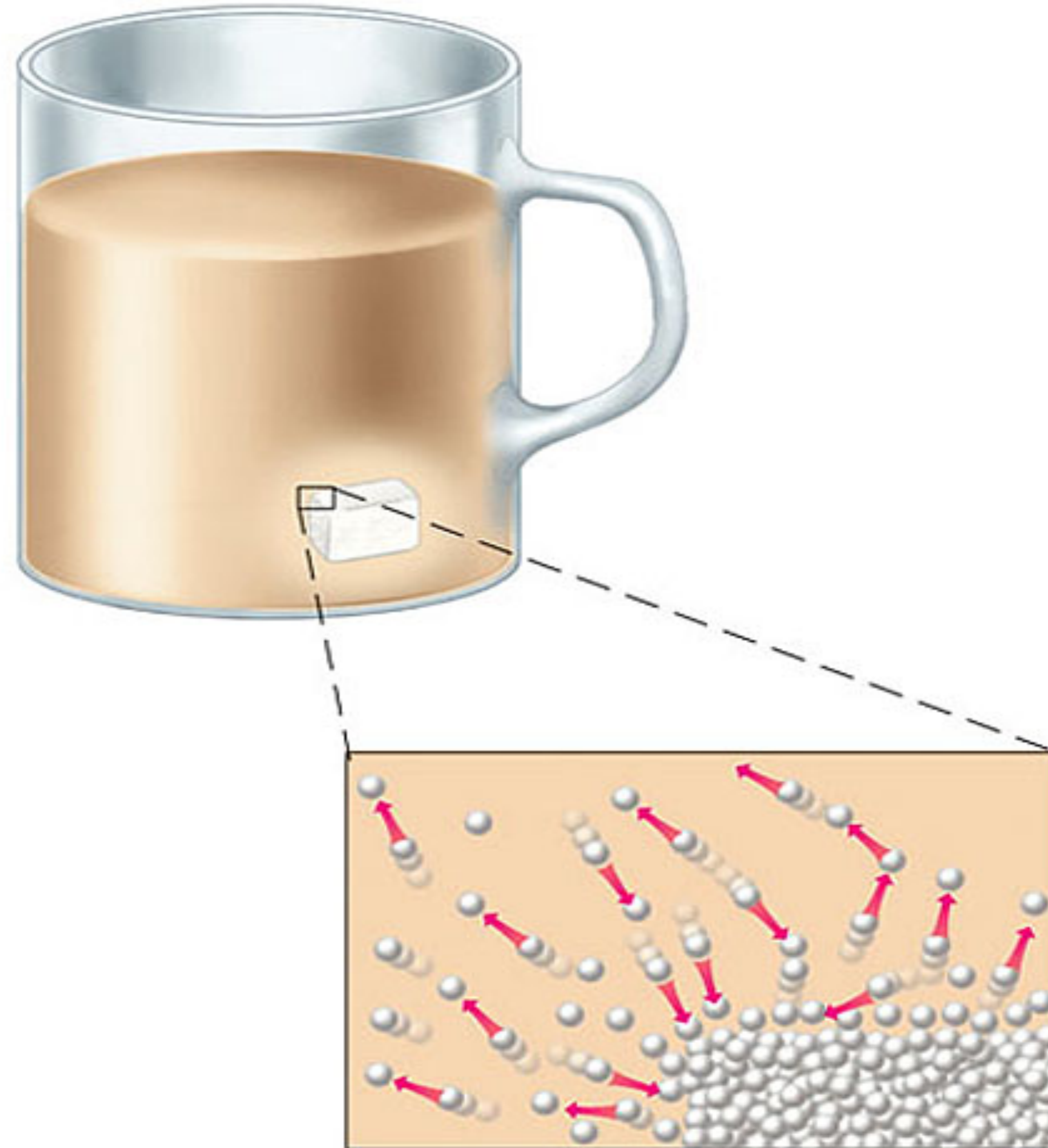
→ Direction of net water movement



Diffusion

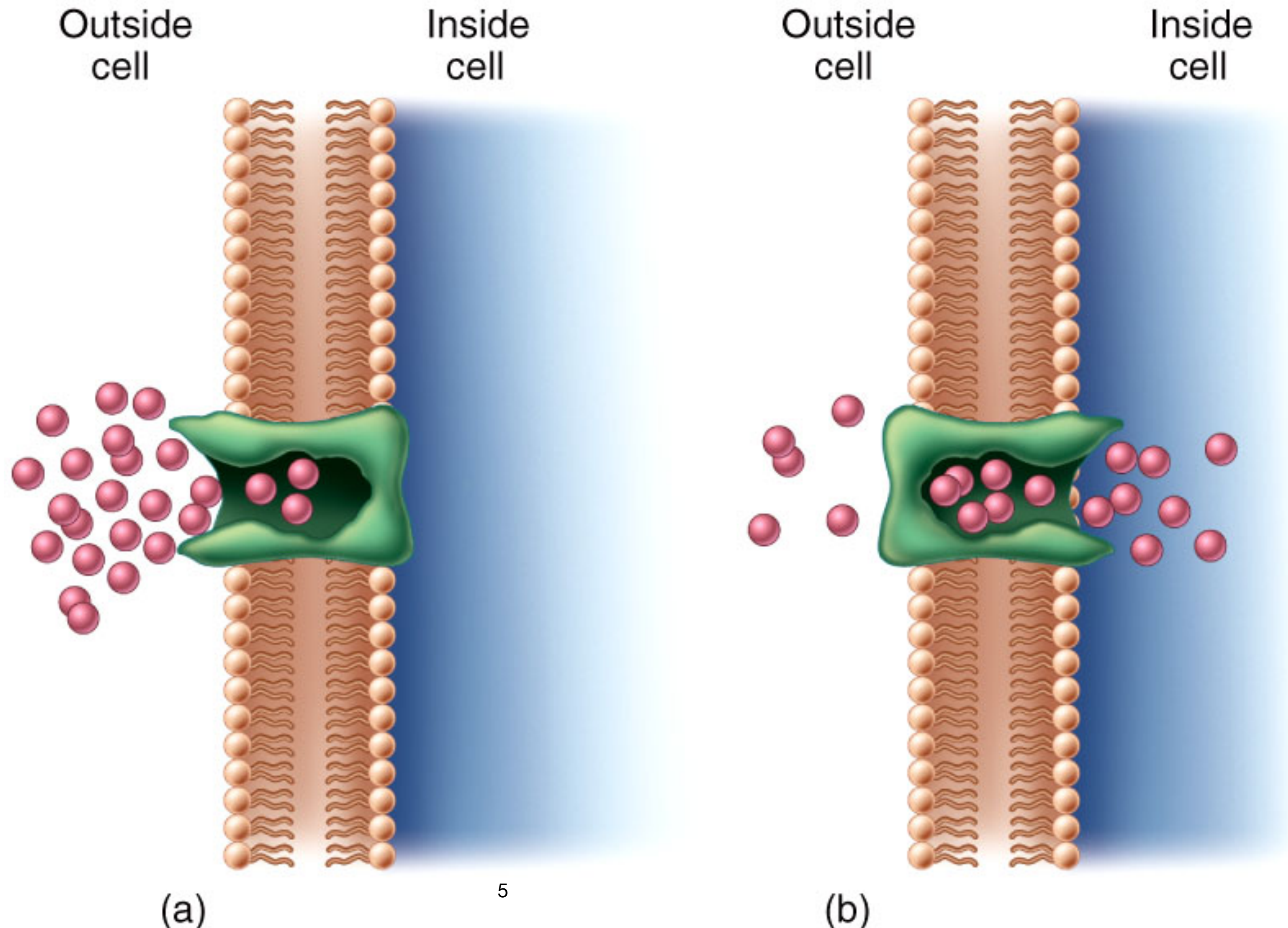
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How Molecules Diffuse in Aqueous Solutions



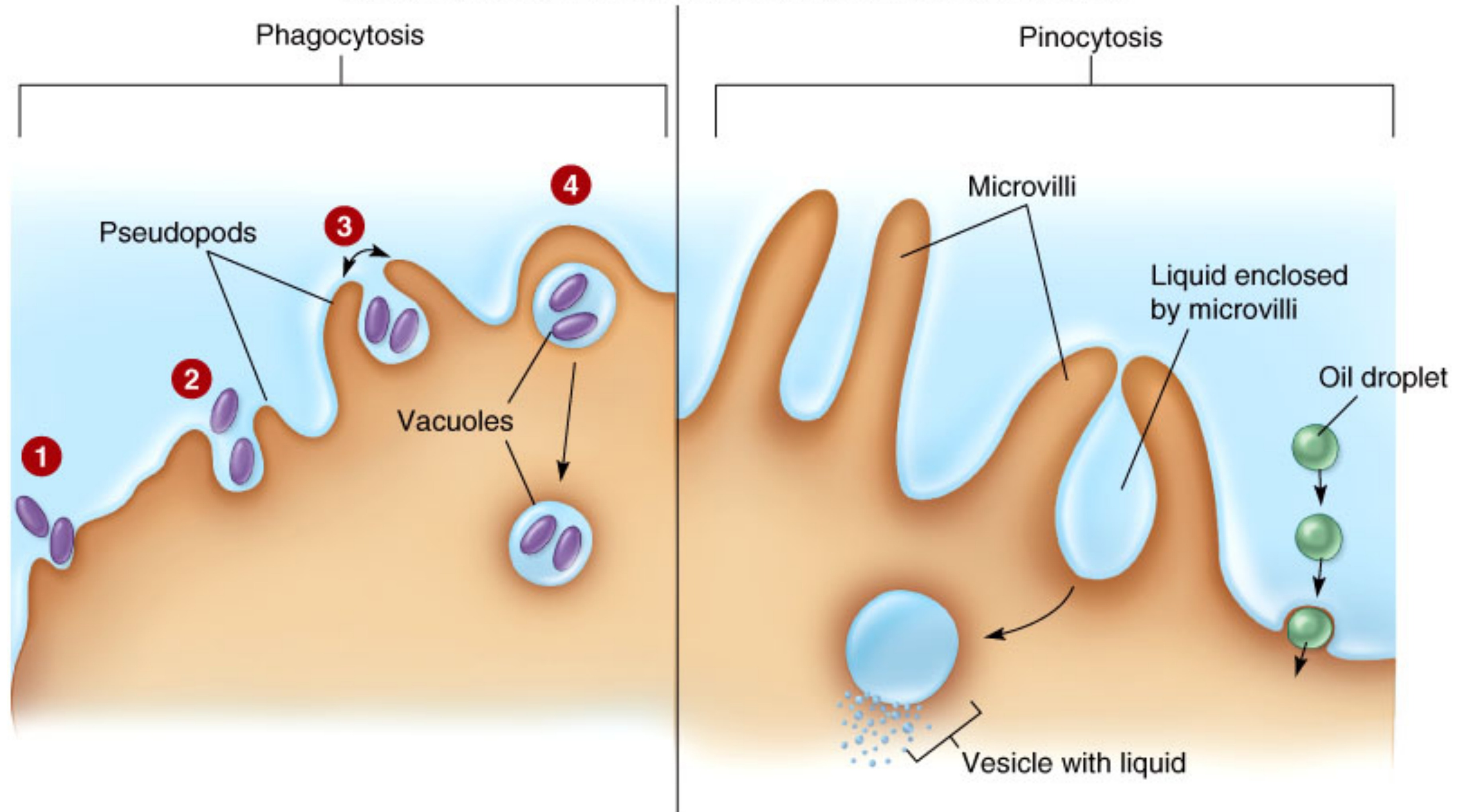
Facilitated diffusion

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Endocytosis

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Active transport

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